

MIDTLAEGER, Norway

WMO: 014330

Lat: 59.833N

Lon: 6.983E

Elev: 1081

StdP: 89.00

Time Zone: 1.00 (EUC)

Period: 01-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.0	-13.6	-23.7	0.5	-6.6	-20.5	0.7	-8.3	18.1	-9.4	16.5	-10.2	6.4	80	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	18.5	11.8	16.7	11.2	15.1	10.6	12.9	16.7	12.1	15.6	11.3	14.6	4.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.4	14.4	10.3	8.9	13.8	9.4	8.4	13.1	39.4	16.8	37.3	15.6	35.3	14.6	14.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.2	14.0	12.1	DB	-16.9	21.6	1.5	1.4	-18.0	22.6	-18.9	23.4	-19.7	24.1	-20.9	25.1	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-5.6	-5.9	-3.9	0.0	3.2	7.0	10.5	9.6	6.3	1.8	-1.6	-4.1
	DBStd	6.63	3.84	3.93	3.96	3.45	3.35	3.03	3.22	3.31	2.95	3.50	3.49	3.74
	HDD10.0	3213	484	446	432	301	212	99	32	51	116	256	348	436
	HDD18.3	6154	743	679	691	551	469	341	242	272	361	513	598	694
	CDD10.0	102	0	0	0	0	1	9	49	37	5	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.0	6.4	5.4	5.6	4.6	4.1	3.8	4.0	4.3	4.8	5.1	5.9	5.6
Precipitation	PrecAvg	1797	226	172	164	90	79	85	97	123	156	205	198	208
	PrecMax	2484	495	390	499	186	211	179	159	239	281	444	464	424
	PrecMin	1153	18	10	20	38	5	15	31	29	31	42	34	6
	PrecStd	325	128	115	100	37	47	45	31	47	76	104	109	126
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	4.4	6.4	9.0	13.6	18.8	21.5	19.7	15.8	11.8	8.1	4.2
		MCWB	1.1	1.3	2.2	4.5	6.8	10.7	13.1	12.2	10.2	7.6	4.8	1.1
	2%	DB	1.2	1.7	3.7	7.3	11.2	15.8	19.2	17.5	13.4	9.0	6.3	2.5
		MCWB	0.3	-0.4	0.5	3.5	6.3	8.5	12.3	11.7	9.5	6.5	4.0	0.6
	5%	DB	0.2	0.5	2.2	5.9	9.7	13.7	17.6	15.9	11.8	7.8	4.7	1.6
		MCWB	-0.5	-0.9	0.3	2.9	5.5	7.9	11.7	11.5	8.9	5.8	3.2	0.1
	10%	DB	-0.6	-0.6	0.9	4.6	8.1	12.0	15.8	14.4	10.5	6.6	3.1	0.6
		MCWB	-1.3	-1.6	-0.5	1.9	4.6	7.1	11.0	11.0	8.2	4.9	1.6	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.3	2.9	5.7	8.4	11.6	14.0	13.3	11.4	8.9	5.7	2.7
		MCDB	2.0	3.8	5.1	7.3	11.7	16.2	19.0	16.9	13.7	10.8	7.3	3.2
	2%	WB	0.5	0.5	1.5	4.3	7.3	10.1	13.2	12.7	10.4	7.2	4.6	1.6
		MCDB	0.8	1.1	2.8	6.3	10.3	14.5	17.6	15.9	12.7	8.7	5.8	2.0
	5%	WB	-0.4	-0.6	0.6	3.2	6.1	8.7	12.5	11.9	9.5	6.2	3.2	0.6
		MCDB	-0.1	0.1	1.9	5.4	8.8	12.3	16.5	15.2	11.5	7.8	4.5	1.2
	10%	WB	-1.2	-1.6	-0.4	2.3	4.9	7.7	11.7	11.2	8.6	5.1	1.9	-0.4
		MCDB	-0.8	-0.8	0.8	4.2	7.7	11.0	15.3	14.3	10.5	6.5	2.9	0.3
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.4	5.0	5.3	5.1	5.9	5.4	4.5	4.3	4.0	3.9	4.1
		MCDBR	4.3	5.2	6.4	6.7	6.4	8.6	8.0	6.2	5.8	5.7	5.0	4.6
		MCWBR	4.2	4.4	4.5	4.0	4.0	4.9	3.8	3.0	3.5	4.4	4.2	4.4
	5% WB	MCDBR	4.2	4.7	5.4	5.5	5.5	7.4	7.1	5.7	5.0	4.9	4.8	4.4
		MCWBR	4.2	4.4	4.4	3.7	3.9	4.6	3.7	3.4	3.7	4.4	4.5	4.5
Clear-Sky Solar Irradiance	taub	0.207	0.243	0.271	0.301	0.306	0.304	0.318	0.312	0.288	0.268	0.218	0.181	
	taud	2.168	2.278	2.254	2.234	2.328	2.486	2.477	2.486	2.549	2.476	2.221	2.044	
	Ebn at Noon	624	793	872	901	923	930	907	887	859	762	597	497	
	Edh at Noon	40	64	91	113	111	97	96	88	69	54	37	29	
All-Sky Solar Radiation	RadAvg	0.31	0.90	2.19	4.02	4.71	4.94	4.53	3.55	2.27	1.09	0.40	0.20	
	RadStd	0.07	0.23	0.46	0.75	0.83	0.61	0.60	0.31	0.24	0.18	0.08	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page